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A cross-platform rumor detection framework considering data privacy protection and different detection capabilities of online social platforms

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ABSTRACT

The anonymity and widespread popularity of online social platforms (OSPs) allow users to share uncertain posts freely, leading to numerous rumors. Similar rumors spread widely across OSPs, resulting in frequent cross-platform rumors (CPRs). Owing to the unique nature of the cross-platform spread, the dual challenges of data privacy protection constraints and differences in the data and detection capabilities of OSPs exacerbate the difficulty of CPR detection. Thus, to detect CPRs effectively, we designed and implemented a novel deep learning framework named Cross Platform Rumor Detection based on Improved Federated Learning (CPRDIFL), which integrates and improves federated learning and the pre-trained Masked and Contextualized BERT (MacBERT). Our framework uses FL to analyze data from OSPs independently, thus avoiding the need for data integration and ensuring the data privacy protection of OSPs. Moreover, MacBERT is deployed on the clients of CPRDIFL to extract contextual features from posts and dynamically update local weights based on the data and detection performance. Weight parameters are dynamically shared between clients and servers and between clients to achieve complementary advantages across OSPs. Our framework was used in six comprehensive experiments in different scenarios, and the experimental results showed that it achieved the best results in CPR detection. This study not only provides an effective solution for CPR detection but also marks a significant step toward the automated detection of cross-OSP information pollution.

1. Introduction

Online social platforms (OSPs) are playing an increasingly important role in facilitating the free flow and wide dissemination of information [1], enabling users to share their lives, express opinions, and exchange ideas across diverse platforms effortlessly [2]. However, the quality of OSP information varies significantly, with large differences in accuracy, objectivity, completeness, and reliability [3], leading to the proliferation of online rumors. Online rumors, also known as online gossips, are unverified information and hot topics that trigger heated public discussions [4]. A similar concept to online rumor is fake news, which is false information that can be verified and the detection of which focuses on determining whether it is true or false [5,6]. Because of the unverified characteristics of online rumors, their public discussions are often disorderly, which increases the risk of negative public opinion and affects the decision-making process of stakeholders [7], posing a serious threat to the credibility of social media and the stability of democratic societies [8]. Therefore, real-time and effective online rumor detection is crucial. It is precisely because of the aforementioned characteristics of online rumors that the detection of them should pay more attention to their dissemination characteristics, especially across different OSPs. Manual fact-checking of rumors is time consuming, laborious, and inefficient [9] and automatic rumor detection based on machine learning (ML) or deep learning (DL) has attracted extensive attention from academia and industry [10].

Leveraging the openness of social platforms, it is common for the same user to have different accounts on various OSPs [11], which is referred to as a “co-user”. While enjoying the convenience of obtaining diversified information [12], such co-users also enable rumor events to spread in a wider cascade across various platforms [13], which are referred to as “cross-platform rumors” (CPRs). Fake news typically remains unverified during its initial emergence, therefore, its cross-platform dissemination dynamics demonstrate parallels with CPRs in the early propagation phase. Given that fake news is typically fabricated with deliberate or even malicious intent to mislead audiences [14–16], it faces heightened supervision. Upon being verified as false, robust

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supervision mechanisms effectively curtail both the spatial scope and temporal persistence of its dissemination. Consequently, its cross-platform spread will exhibit significant mitigation. CPRs are typically unverified and evolve dynamically during dissemination among different OSPs. In particular, collaborative filtering recommendation algorithms for OSPs tend to recommend relevant content to interested users [17]. If the interested users are co-users and the recommended content happens to be CPRs, the speed and scope of the spread of such CPRs will be further increased. It is possible to convince co-users of CPRs further when they have similar information on CPRs from various platforms. For example, the exaggeration and misinterpretation of a Swiss study on COVID-19 vaccines is a typical CPR. The study reported that less than 3 % of people briefly had a slightly elevated blood level of a protein after a COVID-19 booster, which could be a marker of heart injury [18]. Although the original study emphasized that no definitive conclusion could be drawn, the findings were distorted across multiple OSPs, fueling viral posts. Florida Surgeon General Joseph Ladapo claimed on X (formerly Twitter) that the study "proved" COVID-19 boosters cause heart damage; an Instagram user posted a Gateway Pundit headline reading "KILL SHOT: 1 in 35 Moderna recipients show signs of heart injury"; meanwhile, a YouTube user described the study's findings as "off the scale". Despite the absence of such claims in the original study, these misleading rumors spread rapidly across OSPs, triggering widespread public doubt and anxiety about vaccine safety. As CPRs can have a significant impact and serious consequences, effective detection is urgently required for all OSPs.

OSPs can be divided into general and specialized OSPs, which differ significantly in terms of the data volumes and domain-specific characteristics. As a general OSP, Weibo encompasses various domains. According to the 53rd Statistical Report on China's Internet Development [19], as of December 2023, China had 1092 million internet users. Among them, Weibo had 598 million monthly active users (MAUs), representing diverse interests and backgrounds and producing a broad range of content. In contrast, Meiyou, which is a specialized medical and health OSP, reported 40 million MAUs during the same period, primarily focusing on smart diagnoses and treatments. These disparities indicate that general OSPs with large and diverse user bases tend to generate substantial amounts of data across multiple domains [20]. Conversely, specialized OSPs with relatively smaller and more focused user bases produce smaller volumes of high-quality domain-specific content [21]. Such differences in data volume and domain positioning focus directly affect the rumor detection performance of OSPs.

Under the constraints of regulatory strategies, each OSP typically deploys corresponding rumor detection algorithms to realize targeted rumor detection in combination with its own domain characteristics [22]. However, rumor detection by OSPs independently may not solve the problem that rumors that have been detected on one OSP may still be spread on other OSPs because the OSPs may be oriented toward different domains, have different data, and deploy different algorithms. Therefore, CPR detection requires comprehensive consideration of the data and rumor-detection capabilities of each OSP. Previous rumor-detection methods mainly used the direct integration of data from different OSPs to detect rumors [23]. Limited by data privacy protection policies and differences in the performance of detection algorithms of various OSPs, CPR detection inevitably encounters the following challenges:

First, data privacy protection policies allow the data generated by each OSP to be managed and operated by each OSP [24], which limits the flow and sharing of data between OSPs. Under such constraints, CPR detection by directly integrating data from various OSPs is not feasible; this is the first barrier to achieving CPR detection. Second, the differences in domain positioning inevitably lead to asymmetry of the data and rumor detection capabilities. Even if existing effective detection models are deployed on different OSPs, their performance is affected by these differences, and different models are deployed on different OSPs in the real world. The asymmetry in data and rumor-detection capabilities among OSPs poses another barrier that must be addressed in CPR detection. In light of these challenges, this study aims to address the following research questions:

RQ1: How can the bottleneck of data interconnection be overcome under the constraint of data privacy protection of OSPs to realize effective CPR detection?

RQ2: How can we balance the differences in data and detection capabilities among OSPs under the premise of data privacy protection, achieve complementary advantages, and improve the performance of CPR detection?

To address the above questions, we propose a novel framework named Cross Platform Rumor Detection based on Improved Federated Learning (CPRDIFL), which integrates an improved federated learning (FL) framework with the Masked and Contextualized BERT (MacBERT) model to detect CPRs effectively. First, FL, which does not rely on centralized data for model training [25], is selected to train independent rumor-detection models. The models are integrated at the parameter level to achieve CPR detection under the constraint of data privacy protection. Second, we improve the parameter aggregation strategy of the FL framework. The parameters of the local model are dynamically adjusted and aggregated in the global model to update its parameters. The data and capability differences between different OSPs are balanced while ensuring data privacy, and the overall detection performance of CPRDIFL is further improved. Third, MacBERT, as a text correction model with whole word masking and an N-gram masked language model [26], is selected as the local model of CPRDIFL owing to its excellent transfer learning capabilities, contextual understanding, and high adaptability to Chinese datasets to strengthen the CPR detection performance of CPRDIFL further.

The contributions of our study are threefold. (i) We propose CPRDIFL, an innovative CPR detection framework based on improved FL and MacBERT. This framework overcomes the limitations of independently deployed rumor-detection models for individual OSPs and effectively addresses the challenge of leveraging data from multiple OSPs for CPR detection under data privacy protection constraints. (ii) The improved dynamic parameter aggregation strategy in CPRDIFL balances the differences in the data and detection capabilities of multiple OSPs, thereby achieving the complementary and collaborative advantages of multiple OSPs and further enhancing the CPR detection performance. (iii) To the best of our knowledge, few studies have been conducted on CPR detection. Our study is the first to introduce FL into CPR detection and improve the dynamic parameter aggregation strategy of FL, which not only breaks through the bottleneck of CPR detection but also expands and deepens the research and application of FL, offering a new paradigm for intelligent decision support in CPR detection.

The remainder of this paper is organized as follows: Section 2 provides an overview of the relevant literature. Section 3 describes the proposed CPRDIFL framework. The performance of CPRDIFL was evaluated through six experiments, as described in Section 4. Management implications are discussed in Section 5. Finally, the conclusions of this study and future work are presented in Section 6.

2. Related literature

Our research relates to two principal streams within the rumor detection literature, focusing on single-OSP rumor detection and multi-domain/multi-OSP rumor detection, respectively. Given the similarities between fake news, particularly during its early propagation phase, and rumors, this section also incorporates representative relevant literature on fake news detection. Furthermore, we review FL-related studies to provide a robust framework to support the objectives of this research.

2.1. Rumor detection on OSPs

2.1.1. Single-OSP rumor detection

Most rumor-detection studies rely on a single OSP or domain dataset to analyze the rumor content by extracting information features [27].

Research on single-OSP rumor detection can be divided into two primary types. The first category employs ML methods to classify rumor data. For instance, Mahir et al. [28] conducted a comparative analysis of five prominent ML methods, including the support vector machine (SVM), naive Bayes (NB), logistic regression (LR), and other models, to evaluate their efficacy in detecting fake news on Twitter. Capuano et al. [29] analyzed the existing research on content-based fake news detection and found the most performing models to be NB and Multilayer perceptron (MLP). However, these ML-based rumor-detection approaches are data driven and require substantial mass-labeled data for training, which may not always be available across all OSPs. In addition, these methods require manual feature engineering, which is susceptible to subjective biases, rendering them more suitable for short text formats. Fortunately, the BERT series, which is based on transfer learning, offers the advantage of automatically learning feature representations from data and leveraging contextual information. It can be fine-tuned to achieve satisfactory results for specific tasks with minimal labeled data [30]. For example, Kaliyar et al. [31] developed the FakeBERT model to identify political fake news on OSPs effectively during the 2016 US presidential election. Similarly, Babaian and Xu [32] fine-tuned four BERT-based models to identify relevant medical symptoms in spoken language using COVID-19-related data from the CLQ dataset.

2.1.2. Multi-domain and multi-OSP rumor detection

With an increasing number of OSPs containing multi-domain information, multi-domain and multi-OSP rumor detection has gradually become a research hotspot. Existing studies have generally integrated data from various sources using data integration techniques to train models for effective rumor detection [33]. For example, Kim et al. [34] used a dataset containing five rumor domains to train ML-based ensemble methods for improved rumor detection. In another study, Kaur et al. [35] used rumors collected from three public datasets originating from different OSPs and employed 12 DL classifiers to identify fake news across different platforms. Faustini et al. [36] performed a comparison of fake news detection in three languages and two OSPs (Twitter and Weibo), and found that SVM and Random Forest (RF) performed better. Yuan et al. [23] further demonstrated the efficacy of multi-domain fake news detection models by integrating data of eight domains and 12 events from Weibo and Twitter to train and evaluate their proposed model. Li et al. [37] proposed a two-stage model in which BERT model is used to identify rumor texts, evaluated on three datasets collected from Twitter and Weibo. However, data integration methods face significant limitations in addressing the challenges posed by data privacy protection constraints across OSPs, which necessitates further research to explore effective solutions for CPR detection.

2.2. Application of FL

As shown in Fig. 1, FL enables collaboration through information exchange between different clients and a central server without directly integrating client datasets [25], and offers a solution for model training across distributed devices while maintaining data privacy [38]. FL has been widely applied in critical domains such as healthcare and finance to address data privacy challenges by enabling collaborative yet private data usage [21]. In the medical domain, FL enables secure sharing of medical data among various healthcare institutions, thereby safeguarding the data privacy of these organizations during diagnostic and treatment processes [39]. Bardhan et al. [40] used FL to investigate the impact of single versus multiple provider hospital systems on patient prognostic outcomes. Liu and Bi [41] employed FL to mitigate the knowledge disparities among doctors in multiple hospitals by leveraging historical patient data. In the financial domain, FL provides a robust

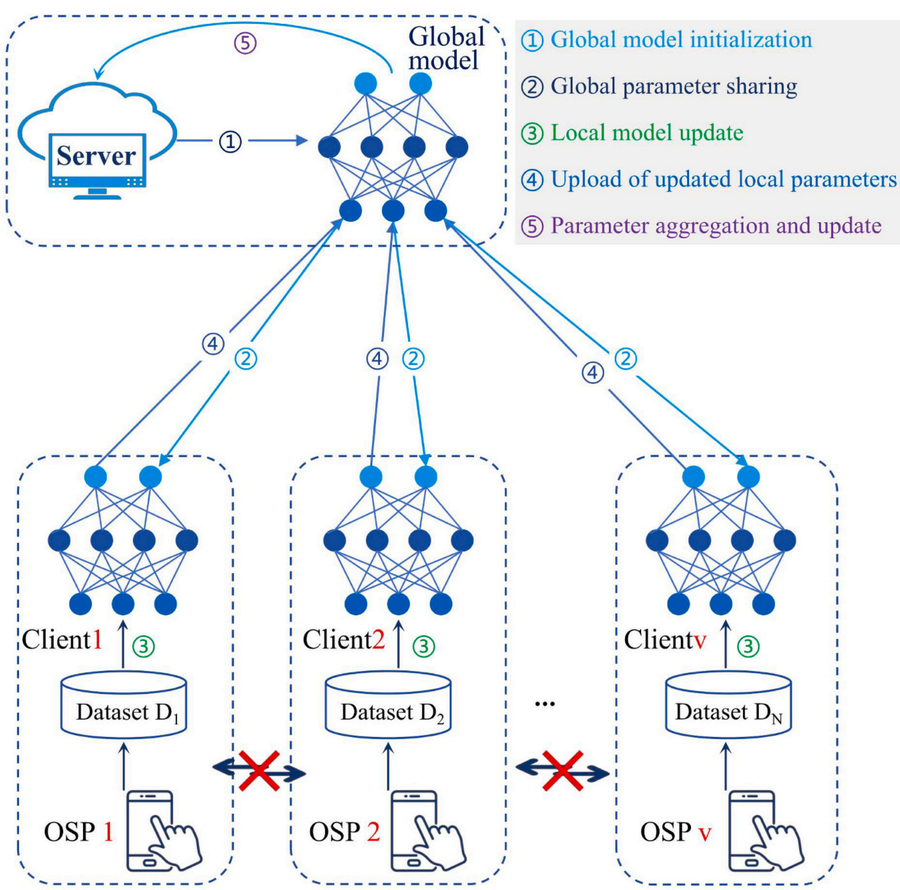

Fig. 1. The framework of federated learning.

framework that allows institutions to share client credit parameters without exposing sensitive client data, which is crucial for building credit-scoring systems and detecting financial fraud [42]. For instance, Wang et al. [43] proposed the FedKT method to solve the issue of asymmetric information in borrower profiles, which hampers accurate creditworthiness assessments. He et al. [44] designed an assessment model for credit scoring based on FL to address the challenges of preserving data privacy in online lending platforms.

In summary, FL effectively mitigates information or data asymmetry in many domains while ensuring the privacy of institutional data. This underscores the significant potential of the FL framework for address the dual challenges of CPR detection mentioned in the Introduction. Inspired by these studies, this work aims to explore the integration of the FL framework into secure and efficient CPR detection.

2.3. Research gaps and questions

Several important research gaps were identified based on a comprehensive analysis of the existing literature. Current studies on single-domain (OSP) rumor detection have primarily focused on data within a single OSP. However, because of the inability to differentiate between various data sources effectively, domains, and detection capabilities of different OSPs, such methods have been shown to be inadequate for detecting rumors across multiple OSPs or domains. Existing multi-OSP and multi-domain rumor-detection methods that utilize data integration cannot address the data privacy constraints in real-world OSPs, resulting in an inability to solve the data privacy protection problem in CPR detection effectively. Therefore, there are currently no effective CPR detection methods. In addition, although the FL framework has been widely applied in domains such as healthcare and finance to address data privacy protection, its application to rumor detection is yet to be fully explored. In particular, dynamic balancing the differences in the detection capabilities of various OSPs, which are caused by differences in the data volume, domains, and models, has not been effectively addressed. This issue is further compounded by the dynamically evolving characteristics exhibited by rumors during cross-platform spread. In summary, existing methods fail to address the dual challenges of CPR detection: data privacy protection across OSPs and the differences in data and detection capabilities across different OSPs. This highlights the need to explore and develop an effective CPR detection framework.

3. Research design

In this section, we first provide a formal representation of the CPR detection problem, and then elaborate on the proposed CPRDIFL framework.

3.1. Problem statement

To clarify the research problems of this study, we draw on previous research on rumor detection and provide a precise mathematical definition of CPR detection as a binary classification problem. Suppose that each entry in the dataset consists of three main parts: the post (c), user who posted the post (u), and OSP on which the post was posted (p). The set of OSPs involved in the spread of CPRs can be defined as.

$P = \{p_1, \cdots, p_i, \cdots, p_v\}, i = 1, 2, \ldots, v$ of OSPs involved in the CPR spread. A user on p_i may post multiple posts, which means that the number of users is much smaller than the number of posts. To facilitate the problem description and reflect the real data structure in OSPs, the following representations of users and posts within OSP p_i are provided without affecting the construction of the model and subsequent experimental results:

$$U_i = \left\{u_i^1, \cdots, u_i^j, \cdots, u_i^{m_i}\right\}, j = 1, 2, \ldots, m_i \tag{2}$$

$$C_i = \left\{c_i^1, \cdots c_i^j, \cdots, c_i^{m_i}\right\}, j = 1, 2, \ldots, m_i \tag{3}$$

where u_i^j and c_i^j denote a certain user and his/her post, respectively, m_i is the number of users or their posts on OSP p_i, and $n = \sum_{i=1}^{v} m_i$ is the number of users or posts on all OSPs. Then, the set of labeled data for training (training and validation sets) on OSP p_i can be defined as

$$E_i^l = (p_i, U_i, C_i, Y_l), y \in Y = \{0, 1\} \tag{4}$$

where Y_l is the label set of posts and y is 0 or 1, indicating that the post is a rumor or not, respectively. Accordingly, the posts that must be detected on OSP p_i can be defined as

$$E_i = (p_i, U_i, C_i). \tag{5}$$

The goal of this study is to learn a model f to detect each post on any OSP p_i.

$$f\left(E|E_i^l\right) \rightarrow Y \tag{6}$$

where $E = (P, U, C)$, $U = \{U_1, \cdots U_i, \cdots, U_n\}$, and $C = \{C_1, \cdots C_i, \cdots, C_n\}$.

To achieve this, two bottlenecks must be overcome. First, the data in E cannot be integrated through the interconnection of OSP p_i owing to data privacy protection. Second, there is an adverse impact of the imbalance of data and capabilities of the OSP on the detection performance under data privacy protection.

3.2. Design of proposed CPRDIFL framework

To address the dual challenges faced by CPR detection, we propose the novel CPRDIFL framework that realizes the complementary advantages of each OSP and improves the CPR detection performance. The CPRDIFL framework is illustrated in Fig. 2.

According to the FL mechanism, CPRDIFL consists of a server and two types of clients: specialized and general clients. Considering the advantages of MacBERT mentioned in the Introduction, it is used as the local model for clients to capture local semantic features. During the training phase, the server first distributes the initial parameters (mainly the model weights) of the global model to all clients. Each client then fine-tunes its deployed MacBERT using labeled posts from each OSP to obtain the updated parameters of the local models while adhering to the data privacy protection constraints. Thereafter, the updated parameters of the local models are provided to the server separately. On the server, the updated local model parameters are aggregated using dynamic coefficients to obtain the updated global model parameters. The updated global model parameters are then redistributed back to each client to update the local model parameters, which can help to mitigate the influence of the data and capability asymmetry of different OSPs on the CPR detection performance. The parameters of both the local and global models are dynamically and continuously updated and iterated until the global model converges. Thus, the CPRDFL framework can be used to detect CPRs.

3.2.1. Local model

The MacBERT model is a Transformer-based pretrained DL model that is designed to learn language representations from a Chinese text corpus. It improves BERT using a refined masking strategy that replaces words with semantically similar words to enhance the contextual understanding. The architecture is shown in Fig. 3.

The encoder in MacBERT consists of multiple repeated modules, each of which contains two main sub-layers: multi-head self-attention and feed-forward neural networks, and each sub-layer is followed by a residual connection and layer normalization (Add&Norm). Each input post x is fed into MacBERT as an input to obtain its embedding $E(x)$. The contextualized text representation obtained by processing $E(x)$ can be defined as

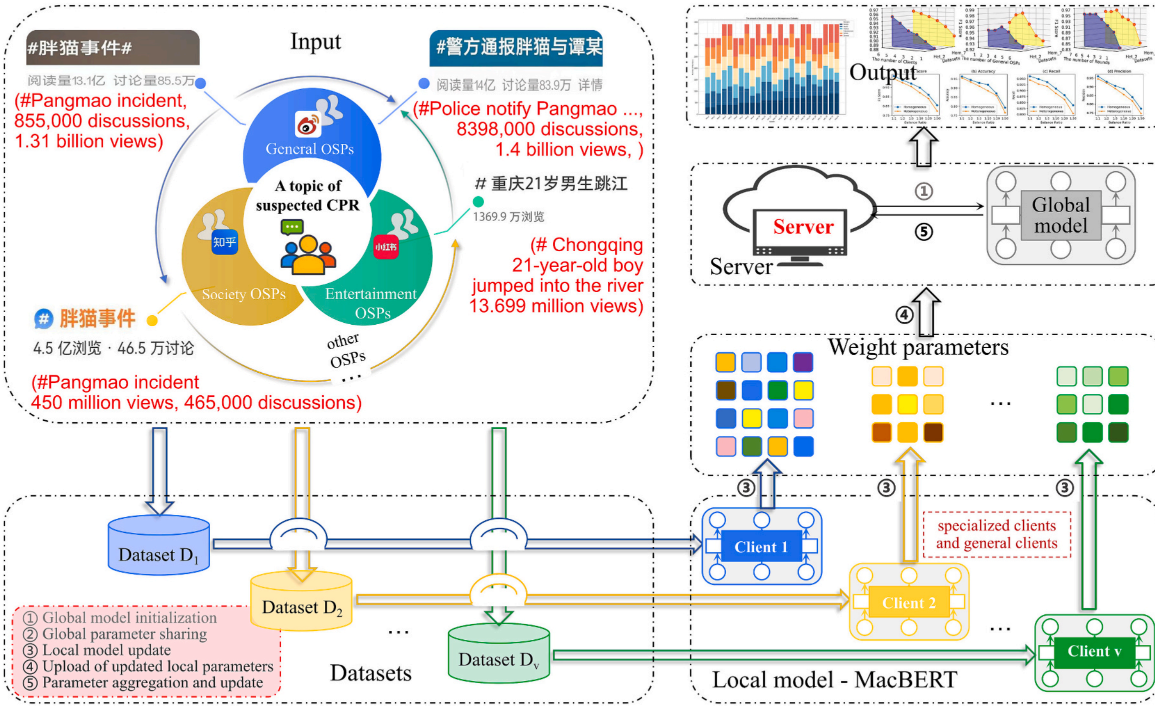

Fig. 2. The overview of CPRDIFL framework.

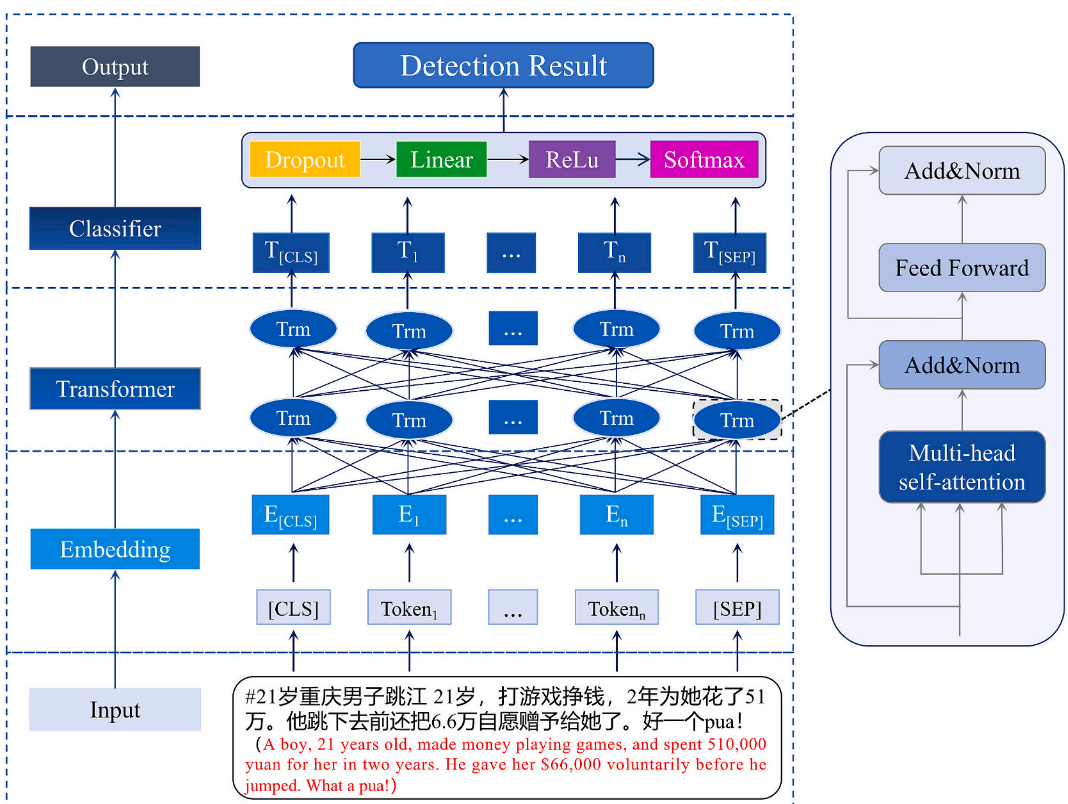

Fig. 3. The network architecture of MacBERT.

$$T(x) = FFN(E(x)) \tag{7}$$

where $T(x)$ is the feature representation of post x and $FFN(\bullet)$ is the feed-forward neural network. A dropout layer, linear layer, and rectified linear unit (ReLU) activation function are added to learn more complex feature representations. Finally, the softmax activation function is added to predict the type of post x with its predicted probability score, which can be defined as

$$\widehat{y}_x = w \bullet T(x) + b \tag{8}$$

where w is the Chinese pre-training weight of MacBERT, b is the bias, and $\hat{y}_x$ is the predicted probability score of post x. If $\hat{y}_x > \alpha$, post x is classified as a CPR. Subsequently, the loss of MacBERT $\mathscr{L}(\mathrm{w})$ is calculated using the cross-entropy loss function, which can be represented by

$$\mathscr{L}(\mathrm{w}) = -\frac{1}{m}\sum_{x=1}^{|E^l|}[y_x\log(\hat{y}_x) + (1-y_x)\log(1-\hat{y}_x)] \tag{9}$$

where y_x is the ground-truth label of post x. Finally, the gradient of *l*oss with respect to each weight is calculated using a back-propagation algorithm, and the weight w is updated to w'. The updated part of the weight parameters of each local model can then be defined as

$$\Delta w_i = w_i' - w \tag{10}$$

where w_i' denotes the updated weight parameters of client p_i.

3.2.2. Global model

In CPRDIFL, the server aggregates updated weight set $\Delta W = \{\Delta w_1, \cdots \Delta w_i, \cdots, \Delta w_v\}$ received from all clients to update the weight parameters of the global model. The key to training the global model and ensuring its convergence lies in efficiently minimizing the objective function. To improve the generalization performance to adapt to the complexity of CPR propagation, we introduce a dynamic coefficient β into CPRDIFL for the interaction between the server and clients to mitigate the asymmetry caused by differences in data and detection abilities among various OSPs. First, a dynamic regularization term related to β is constructed in the objective function of the local models, which can reflect the weight contributions based on the data and detection capabilities of different OSPs while effectively preventing model overfitting. Second, when updating the global model weight parameters, we also account for β to balance the differences in data and detection capabilities among OSPs. The local objective function of client p_i and the global objective function can respectively be expressed as

$$f_i(w) = \mathscr{L}(\mathrm{w}) + \frac{\beta_i}{2}\|w_i - w_g\|^2 \tag{11}$$

$$F(w) = \frac{1}{n}\sum_{i=1}^{v} f_i(w) \tag{12}$$

where w_i and $\mathscr{L}(\mathrm{w})$ are the weight of the local model and updated loss function of client p_i respectively, w_g is the weight of the global model, and β_i is the dynamic coefficient of client p_i, which is dynamically adjusted according to its local data and model capability.

In training round $k(k = 1, \cdots k, \cdots, s)$, the server sends the weight parameters of global model w_g^k to client p_i. Then, client p_i uses a subset e_i^l of E_i^l to train the local model and update its weight, which can be represented by

$$w_i^{k+1} = w_g^k - \eta\sum_{k=1}^{s}\nabla_w\mathscr{L}(w_i; e_i^l) - \eta\beta_i\left(w_i^{k+1} - w_g^k\right) \tag{13}$$

where w_g^k is the updated model weight of client p_i in round $k+1$, η is the learning rate, and $\nabla_w\mathscr{L}(\mathrm{w}; e_i^l)$ is the gradient of $\mathscr{L}(\mathrm{w})$ with respect to w_g^k on subset e_i^l of dataset E_i^l. Then, the updated weight parameters of client p_i in round $k+1$ can be represented by

$$\Delta w_i^{k+1} = w_i^{k+1} - w_g^k \tag{14}$$

where Δw_i^{k+1} is the update weight part of client p_i. Client p_i sends Δw_i^{k+1} to the server rather than sharing directly to protect the data privacy of the different OSPs. Considering the differences in the data and detection capabilities of OSPs, we introduce an improved dynamic parameter aggregation strategy to update the weight parameters of global model, which can be defined as

$$w_g^{k+1} = w_g^k + \left(\sum_{i=1}^{v}\left(\frac{m_i}{n}\Delta w_i^{k+1}\Big/\beta_i\right)\right)\Big/\left(\sum_{i=1}^{v}\left(\frac{m_i}{n}\Big/\beta_i\right)\right) \tag{15}$$

where w_g^{k+1} is the global model weight in round $k+1$. The new global weight, w_g^{k+1}, is then distributed to all clients. This process continues until the objective function $F(w)$ is minimized and the global model converges.

3.2.3. CPR detection with CPRDFL

In the test phase, the unlabeled post x is input into CPRDIFL, and the softmax activation function (see Eq. (8)) outputs the probability score $\hat{y}_x$. As $\hat{y}_x$ represents the probability of whether post x is a rumor, $\hat{y}_x$ is compared to a pre-defined threshold α, which is set to 0.5 following a previous study [45]. If $\hat{y}_x > \alpha$, post x is classified as a CPR; otherwise, post x is not a CPR. The detection algorithm is shown in Fig. 4.

4. Experiment

4.1. Datasets

To evaluate the effectiveness of CPRDIFL in CPR detection, we employed the Weibo21 dataset [46], which includes data from various domains. Given the limitations of the dataset, we simulated the data for five specialized OSPs and one general OSP. The posts of specialized OSPs were simulated using data from a single domain, whereas the posts of general OSP were simulated using data from multiple domains. To maintain data diversity and prevent overlap, the posts were randomly sampled without repetition. We developed two types of datasets: homogeneous and heterogeneous. Considering that each dataset was generated through random sampling, the process was performed five times to enhance the credibility of the subsequent experiments.

First, we developed seven homogeneous datasets of varying sizes (Table 1) to simulate a data privacy protection scenario. To avoid the additional impact of differences in data and capabilities across OSPs, each dataset contained post data from six OSPs, with the same number of posts from each OSP set. Second, we developed 24 heterogeneous datasets to simulate the differences in data and capabilities across OSPs while maintaining data privacy protection. Specifically, the OSP contribution orders for the seven homogeneous datasets were obtained (as detailed in Table S.1 of the Supplementary Material). The OSP contribution orders for the Hom_6 and Hom_7 datasets were identical, indicating that only six distinct OSP contribution orders were obtained. Finally, the six OSP contribution orders were combined with two different data sizes (from 300 to 800, totaling 3300, and from 400 to 900, totaling 3900), two matching methods (both forward and reverse) were performed, and 24 heterogeneous datasets were constructed. The 24 heterogeneous datasets with detailed descriptions were shown in Fig. S.1 in Supplementary Material.

4.2. Experimental setup and evaluation metrics

Table 2 specifies the implementation details and hyperparameters not only for the MacBERT but also for other components of the proposed CPRDIFL framework. The optimal hyperparameters of the MacBERT and the baseline models (see Section 4.3) were achieved by conducting preliminary experiments, which were shown in Supplementary material Table S.2 and Fig. S.2. In addition, four detection metrics, including accuracy, precision, recall, and F1 score, were used to evaluate the performance of CPRDIFL.

4.3. Experimental design

We designed six main types of experiments. All experiments were repeated five times with different data, and the average F1 score was

Algorithm 1: The training and test phase of CPRDIFL

Input: The collections of OSPs, users, and posts are::
$P = \{p_1, ..., p_i, ..., p_v\}$, $U_i = \{c_i^1, ..., c_i^j, ..., c_i^{m_i}\}$ and
$C_i = \{c_i^1, ..., u_i^j, ..., u_i^{m_i}\}$

Train ;
`// the number of clients`
for $i = 1$ **to** v **do**
  `// MacBERT encoder`
  $T(x) \leftarrow FFN(E(x))$;
  `// MacBERT predicted probability score`
  $\hat{y}_x \leftarrow w \cdot T(x) + b$;
  `// Optimization with cross-entropy loss function`
  $L(w) \leftarrow -\frac{1}{m}\sum_{x=1}^{E^l}\left[y_x log(\hat{y}_x) + (1 - y_x) log(1 - \hat{y}_x)\right]$;
  `// Weights are updated`
  $\Delta w_i \leftarrow w_i^{'} - w$;
end
`// The server updates the global model weights`
$f_i(w) \leftarrow L(w) + \frac{\beta_i}{2}\|w_i - w_g\|^2$;
$F(w) \leftarrow \frac{1}{n}\sum_{i=1}^{v} f_i(w)$;
`// Local model training updates weights`
$w_i^{k+1} \leftarrow w_g^k - \eta\sum_{k=1}^{s}\nabla_w\ \mathcal{L}\ (w; e_i^l) - \eta\beta_i(w_i^{k+1} - w_g^k)$;
`// Client weights updated`
$\Delta w_i^{k+1} \leftarrow w_i^{k+1} - w_g^k$;
`// the dynamic parameter aggregation strategy update the global weight`
$w_g^{k+1} \leftarrow w_g^k + \sum_{i=1}^{v}(\frac{m_i}{n}\Delta w_i^{k+1}/\beta_i)/\sum_{i=1}^{v}(\frac{m_i}{n}/\beta_i)$;
Test ;
$T(x) \leftarrow FFN(E(x))$;
`// Outputs a probability score`
$\hat{y}_x \leftarrow Softmax(ReLU(Linear(T(x))))$;

Fig. 4. Pseudocode for the training and test phase of CPRDIFL.

Table 1
Statistics of homogeneous datasets.

Dataset name	Number of posts on one OSP	Total number of posts	Dataset name	Number of posts on one OSP	Total number of posts
Hom_1	300	1800	Hom_5	700	4200
Hom_2	400	2400	Hom_6	800	4800
Hom_3	500	3000	Hom_7	900	5400
Hom_4	600	3600			

Table 2
Implementation details and hyperparameters of CPRDIFL.

Component	Implementation Details and Hyperparameters
FL Framework	**Training rounds**: 5; **Clients**: 6; **Batch size**: 8; **Aggregation**: dynamical CPRDIFL with =0.1 proximal term
Local Model	**Model**: MacBERT-base (Chinese pretrained), 12 transformer blocks, 12 self-attention heads, with hidden size 768; **Tokenizer**: BertTokenizer with max_length = 512; **Learning rate**: 1e-5; **Batch size**: 8; **Epoch**: 10; **Dropout**: 0.5
Classifier	**Loss function**: CrossEntropyLoss; **Optimizer**: Adam; **Architecture**: Linear layer (768 → 2) + ReLU; **Activation**: ReLU
Training Configuration	**Python**: 3.10; **Deep learning library**: Pytorch; **GPU**: NVIDIA GeForce RTX 4090 (CUDA 11.8); **CPU**: 13th Gen Intel (R) Core (TM) i9-13900KF; **Data split**: 90 % training (90 % training and 10 % validation) and 10 % test (per domain)

calculated to avoid randomness affecting the results. The purposes and practice of each experiment were as follows:

Experiment 1: Three scenarios were designed for seven homogeneous datasets to evaluate the improvement and CPR detection performance of CPRDIFL under data privacy protection constraints. First, we analyzed the impact of privacy protection constraints on the CPRDIFL detection performance in scenarios with and without privacy protection. Second, a comparison experiment was conducted to evaluate the effectiveness of the dynamic parameter aggregation strategy. We compared 11 methods as baseline models, including six pre-trained BERT-based models that perform exceptionally well in text classification and five methods that have been widely applied in recent rumor-detection studies. **Experiment 2**: To evaluate the performance of CPRDIFL in CPR detection under the dual challenges of data privacy protection and the differences in data and detection capabilities, the top three detection methods from Experiment 1 were selected to detect CPRs in 24 heterogeneous datasets under the three scenarios. **Experiment 3**: To analyze the performance sensitivity of CPRDIFL to different hyperparameter settings, three key hyperparameters (clients, training rounds, and general OSPs) were selected and adjusted to evaluate their effects on the CPR detection performance of CPRDIFL. **Experiment 4**: To evaluate the generalization ability of CPRDIFL, we evaluated its detection capability in varying OSPs, cross-OSP, diverse languages, and different events on four English rumor datasets from different OSPs. **Experiment 5**: We compared CPRDIFL with three other detection methods, namely single-OSP rumor detection, multi-domain rumor detection, and cross-domain rumor detection from several aspects. **Experiment 6**: To analyze the effect of the ratio of rumor to non-rumor data on the performance of CPRDIFL, we established six different ratios. The ratios of rumor to non-rumor were set to 1:1, 1:2, 1:10, 1:20 (the ratio in reality [13]), and 1:50.

4.4. Experimental results

4.4.1. Experiment 1: Analysis of detection performance under data privacy protection

Three scenarios were established to evaluate the impact of data privacy protection limitations and the dynamic parameter aggregation strategy of FL on the CPR detection performance. In Scenario 1, CPR detection was performed using integrated data without considering data privacy protection constraints. In Scenarios 2 and 3, CPR detection was implemented under data privacy protection constraints based on the FL framework with and without the dynamic parameter aggregation strategy, respectively. The experimental results (F1 scores) on the seven homogeneous datasets are shown in Table 3, and detailed results of the four evaluation indicators are shown in Table S.3 of the Supplementary Material.

As shown in Table 3, the performance of the BERT-based models was generally better than that of ML methods such as SVM. The top three models with the best performance across various scenarios were RoBERTa, BERT-wwm-ext, and MacBERT, which may be attributed to the superior architectural optimizations and strong adaptability to Chinese data of these models. In scenarios with data privacy protection, the detection performances of all models based on IFL consistently outperformed those based on FL, with the highest F1 score of 96.67 %, proving the effectiveness of the dynamic parameter aggregation strategy applied to CPRDIFL. In addition, the average performance difference between the six BERT-based models in Scenarios 1 and 2 was 5.15 %, whereas the performance difference of ALBERT between the two scenarios was a staggering 8.99 %. This demonstrates the negative effects of data privacy protection on CPR detection. For Scenarios 1 and 3, the average and maximum performance differences of the BERT-based models were 1.58 % and 2.93 %, respectively. As can be observed from the above analysis, although the dynamic parameter aggregation strategy mitigated the impact of data privacy protection, reducing the smallest performance difference to 0.48 %, the negative influence remained, underscoring the importance of FL in detecting CPRs.

4.4.2. Experiment 2: Detection performance comparison under dual challenges

The top three methods in Experiment 1, namely RoBERTa, BERT-wwm-ext, and MacBERT, were selected to evaluate the detection performance under the dual challenges of CPR detection. The experiments were performed under the three scenarios on 24 heterogeneous datasets, and the experimental results are shown in Table 4, with detailed results presented in Table S.4 of the Supplementary Material. It can be observed that the three methods consistently exhibited strong detection performance in all scenarios, in line with their performance in Experiment 1. However, the simultaneous occurrence of the two challenges exacerbated the performance decline of all methods, with the best F1 score of 95.28 % in CPRDIFL. The dynamic parameter aggregation strategy in CPRDIFL significantly mitigated this decline and CPRDIFL achieved the highest detection performance on 19 of the 24 heterogeneous datasets.

In the case of the heterogeneous datasets, data privacy protection still hindered the CPR detection performance of all methods. The average performance difference between the six BERT-based models in Scenarios 1 and 2 was 2.36 %, and the maximum difference between the two scenarios was 6.56 %. CPRDIFL showed a marked improvement, reducing the average difference to 0.68 %, which proves that our improved FL framework with the dynamic parameter aggregation strategy mitigates the negative impact of data privacy protection constraints. Although some performance fluctuations were observed in the heterogeneous datasets, the F1 scores of the three selected methods based on IFL were better than those based on FL. Furthermore, the 2.32-fold reduction in the performance difference (1.25 % vs. 0.68 %) compared with Experiment 1 demonstrates the effectiveness of the dynamic parameter aggregation strategy in narrowing the performance difference between CPRDIFL and the methods using integrated data on

Table 3
Performance of different methods on homogeneous datasets.

Scenarios	Methods	Hom_1	Hom_2	Hom_3	Hom_4	Hom_5	Hom_6	Hom_7
Scenario 1: No data privacy protection	BERT	88.60 %	90.10 %	91.36 %	93.13 %	94.96 %	95.41 %	96.46 %
	RoBERTa	90.67 %	92.39 %	93.72 %	94.59 %	95.41 %	96.66 %	**97.68 %*****
	ALBERT	85.28 %	87.81 %	89.98 %	92.26 %	92.68 %	93.16 %	94.20 %
	BERT-wwm-ext	92.63 %	93.55 %	94.38 %	95.85 %	96.23 %	96.54 %	**97.49 %****
	MacBERT	93.40 %	94.14 %	95.42 %	96.08 %	96.30 %	96.82 %	**97.33 %***
	DeBERTa	91.87 %	93.30 %	94.37 %	95.04 %	95.85 %	96.14 %	96.67 %
	SVM	85.15 %	86.82 %	87.60 %	87.60 %	89.59 %	89.59 %	89.74 %
	LR	77.70 %	77.59 %	79.36 %	79.88 %	81.59 %	82.33 %	83.92 %
	RF	77.38 %	79.08 %	82.32 %	79.50 %	80.66 %	82.79 %	84.08 %
	NB	76.21 %	80.16 %	81.57 %	85.49 %	86.66 %	87.39 %	87.68 %
	MLP	80.02 %	82.62 %	85.50 %	87.21 %	89.20 %	90.16 %	92.24 %
Scenario 2: considering data privacy protection based on FL	BERT_FL	85.02 %	86.18 %	88.55 %	89.77 %	91.11 %	93.19 %	94.22 %
	RoBERTa_FL	87.07 %	89.56 %	90.56 %	92.99 %	93.91 %	94.42 %	**95.57 %*****
	ALBERT_FL	74.77 %	77.13 %	79.21 %	80.64 %	85.29 %	86.30 %	89.11 %
	BERT-wwm-ext_FL	82.02 %	87.72 %	90.73 %	92.68 %	94.46 %	95.36 %	**94.03 %***
	MacBERT_FL	84.29 %	87.47 %	90.19 %	92.26 %	93.02 %	93.91 %	**94.67 %****
	DeBERTa_FL	81.09 %	83.42 %	86.51 %	87.46 %	89.89 %	91.58 %	92.26 %
	SVM_FL	79.85 %	79.85 %	80.46 %	81.47 %	81.05 %	82.10 %	82.42 %
	LR_FL	52.03 %	53.70 %	56.12 %	56.64 %	58.02 %	60.10 %	61.93 %
	RF_FL	70.03 %	71.44 %	75.75 %	76.98 %	76.36 %	78.83 %	81.98 %
	NB_FL	63.75 %	70.67 %	73.57 %	78.39 %	80.69 %	82.04 %	84.92 %
	MLP_FL	69.96 %	78.28 %	82.07 %	83.68 %	86.54 %	89.11 %	90.89 %
Scenario 3: considering data privacy protection based on IFL	BERT_IFL	86.76 %	88.78 %	90.54 %	92.42 %	94.08 %	94.83 %	95.30 %
	RoBERTa_IFL	89.98 %	91.58 %	92.27 %	93.43 %	94.95 %	95.67 %	**96.67 %*****
	ALBERT_IFL	81.95 %	83.95 %	85.19 %	86.94 %	90.69 %	92.43 %	93.73 %
	BERT-wwm-ext_IFL	88.92 %	90.23 %	91.99 %	92.70 %	94.67 %	95.98 %	**96.22 %****
	MacBERT_IFL (**CPRDIFL**)	91.39 %	92.10 %	94.17 %	94.95 %	95.31 %	95.76 %	**96.10 %***
	DeBERTa_IFL	91.15 %	91.65 %	93.37 %	94.30 %	94.84 %	95.73 %	96.06 %

Notes: ***, **, and * indicate the first-, second-, and third-highest F1 scores, respectively, for each scenario. IFL and FL represent the FL framework with and without the dynamic parameter aggregation strategy, respectively.

Table 4
F1 scores of different methods on heterogeneous datasets.

Dataset	Scenario 1			Scenario 2			Scenario 3		
	Ro BERTa	BERT-wwm-ext	Mac BERT	Ro BERTa_FL	BERT-wwm-ext_FL	Mac BERT_FL	Ro BERTa_IFL	BERT-wwm-ext_IFL	Mac BERT_IFL (**CPRDIFL**)
Het_1	95.82 %	94.89 %	95.36 %	90.46 %	92.20 %	93.24 %	94.49 %	94.56 %	**94.88 %**
Het_2	95.84 %	95.50 %	96.18 %	93.06 %	90.13 %	91.53 %	94.35 %	94.68 %	**95.28 %**
Het_3	92.69 %	92.22 %	92.36 %	90.92 %	91.48 %	90.77 %	**92.60 %**	92.04 %	92.15 %
Het_4	93.80 %	93.33 %	93.50 %	91.91 %	92.13 %	91.82 %	92.50 %	92.75 %	**93.44 %**
Het_5	92.06 %	91.85 %	91.88 %	86.70 %	89.91 %	90.19 %	**91.42 %**	90.57 %	91.02 %
Het_6	95.53 %	94.51 %	94.52 %	91.82 %	92.90 %	92.80 %	94.09 %	93.38 %	**94.50 %**
Het_7	95.61 %	95.23 %	96.12 %	92.12 %	94.27 %	92.15 %	94.25 %	94.85 %	**95.06 %**
Het_8	94.61 %	93.61 %	94.02 %	92.11 %	92.58 %	92.62 %	93.00 %	93.51 %	**93.67 %**
Het_9	93.70 %	93.07 %	93.30 %	91.85 %	91.86 %	90.61 %	92.48 %	92.26 %	**93.25 %**
Het_10	93.83 %	93.90 %	93.62 %	91.95 %	92.41 %	91.65 %	92.42 %	92.44 %	**93.18 %**
Het_11	93.28 %	92.95 %	93.12 %	91.50 %	90.31 %	91.78 %	**92.98 %**	92.48 %	92.78 %
Het_12	94.10 %	93.27 %	93.57 %	91.81 %	91.34 %	91.57 %	**93.26 %**	92.90 %	93.06 %
Het_13	94.63 %	93.99 %	94.38 %	92.59 %	91.15 %	91.76 %	93.26 %	93.53 %	**94.17 %**
Het_14	95.15 %	95.02 %	95.56 %	93.55 %	91.31 %	93.33 %	94.93 %	94.96 %	**95.22 %**
Het_15	95.06 %	94.35 %	94.61 %	92.25 %	91.76 %	92.51 %	93.94 %	93.64 %	**93.98 %**
Het_16	94.50 %	93.84 %	94.60 %	92.46 %	92.62 %	92.77 %	93.99 %	93.71 %	**94.12 %**
Het_17	93.25 %	92.59 %	93.81 %	91.29 %	91.15 %	91.80 %	**92.95 %**	92.07 %	92.53 %
Het_18	91.99 %	92.51 %	92.67 %	91.18 %	90.95 %	91.68 %	91.80 %	92.21 %	**92.61 %**
Het_19	92.43 %	92.13 %	93.65 %	88.38 %	90.58 %	90.63 %	91.24 %	91.95 %	**92.45 %**
Het_20	94.71 %	94.17 %	94.58 %	93.44 %	93.01 %	93.36 %	93.82 %	93.88 %	**94.51 %**
Het_21	93.11 %	92.54 %	94.02 %	89.91 %	90.80 %	90.75 %	92.14 %	92.05 %	**92.85 %**
Het_22	92.35 %	92.49 %	93.58 %	89.56 %	90.84 %	90.96 %	90.94 %	92.23 %	**92.89 %**
Het_23	92.80 %	92.20 %	92.46 %	90.60 %	90.96 %	90.02 %	91.36 %	92.02 %	**92.24 %**
Het_24	90.56 %	92.39 %	93.96 %	84.00 %	87.56 %	89.83 %	90.33 %	91.56 %	**91.85 %**

heterogeneous datasets, thereby proving the effectiveness of CPRDIFL for enhancing CPR detection performance in real-world scenarios.

4.4.3. Experiment 3: Sensitivity analysis of CPRDIFL

We performed three groups of hyperparameter sensitivity analysis experiments and selected the homogeneous dataset Hom_7 and heterogeneous dataset Het_2 to evaluate the impact of the three hyperparameters on the CPR detection performance of CPRDIFL. First, the number of clients was adjusted from one to six, the number of training rounds was fixed at three, and the number of general OSPs was set to one to investigate the effect of changes in the number of federated OSPs on the CPR detection performance. Second, the number of general OSPs was adjusted from one to six, the number of clients was fixed at six, and the number of training rounds was set to three to validate the necessity of having both general and specialized OSPs in CPRDIFL; that is, to evaluate the roles of different types of OSPs in CPR detection. The scenario with six general OSPs was an extreme case of no specialized OSPs. Third, the number of training rounds was varied from one to seven, the number of clients was fixed at six, and the number of general OSPs was set to one to determine the preferred training rounds to avoid overfitting while ensuring that CPRDIFL was fully trained. The experimental results are shown in Fig. 5.

As shown in Fig. 5 (a), the performance exhibited a gradual upward trend for the two datasets as the number of clients increased. For example, when the number of clients increased from one to six, the F1 score improved by 5.62 %. The positive impact of increased client numbers on the detection performance indicates that greater participation and collaboration among OSPs contribute to the CPR detection of CPRDIFL. As the number of general OSPs increased, the performance showed a trend of first increasing and then decreasing, as shown in Fig. 5 (b), indicating that the participation of general OSPs can improve the detection performance; however, too many general OSPs will lead to performance degradation. The experimental results suggest that the large and extensive domain data and high detection ability provided by general OSPs are of great help in CPR detection; however, the professional and in-depth specialized information provided by specialized OSPs is equally essential for CPR detection. Furthermore, as the number of training rounds increased, the detection performance of CPRDIFL improved, as shown in Fig. 5 (c), thereby proving the effective learning capability of CPRDIFL. The performance increased significantly when the number of training rounds increased from one to five, but the performance improvement plateaued or even declined slightly beyond five rounds. One possible reason for this is the overfitting caused by the excessive utilization of training resources.

4.4.4. Experiment 4: Generalization experiments on other public datasets

This experiment was performed on multiple public rumor-detection datasets from various OSPs to evaluate the generalization capability of CPRDIFL across different OSPs and language backgrounds. Three widely used public datasets were selected: PHEME [47], Twitter1516 [48], and

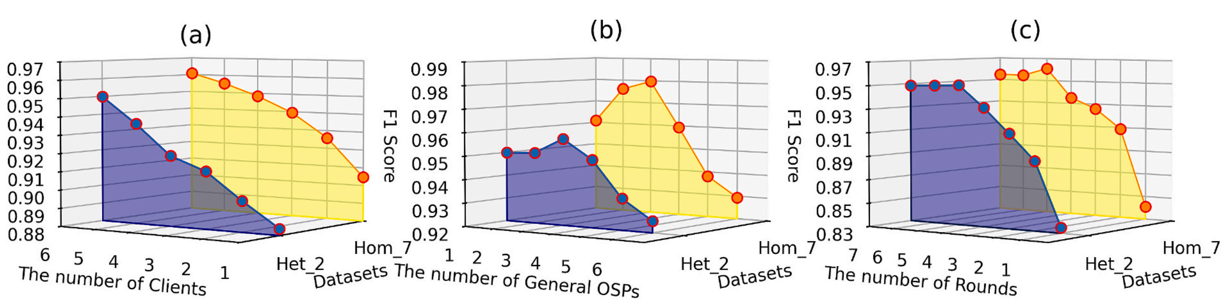

Fig. 5. Performance of CPRDIFL in hyperparameter sensitivity analysis.

Covid19 [49]. These datasets encompass different content sources and event types and provide a diverse set of test scenarios. Rather than using the full datasets, we randomly selected relevant samples by rumor event from each dataset. As observed in Experiment 3, which involved more OSPs and led to better performance, we combined the Twitter1516 and Covid19 datasets to create a mixed dataset to validate the performance of CPRDIFL on the multi-OSP dataset. The experimental results are listed in Table 5.

As shown in Table 5, CPRDIFL achieved the highest F1 score of 90.73 % with an accuracy of 90.70 % on the PHEME dataset, indicating its strong capability for detecting event-specific rumors. The performance on the Twitter1516 dataset, with an F1 score of 81.33 %, was relatively low, likely owing to the brevity and limited context of the tweets. On the Covid19 and mixed datasets, CPRDIFL showed a balanced performance, with F1 scores of 85.85 % and 87.74 %, respectively. By achieving competitive performance on datasets with varying languages, events, and OSPs, CPRDIFL demonstrated robustness and adaptability, further emphasizing its potential for real-world applications.

4.4.5. *Experiment 5: Detection performance comparison with state-of-the-art methods*

In this experiment, we compared CPRDIFL with existing relevant methods, including single-OSP rumor detection, multi-domain rumor detection, and cross-domain rumor detection, to reflect the comparative performance advantages of CPRDIFL. All methods were trained on the Weibo21 dataset. We evaluated their performance from multiple perspectives, such as single OSP/domain detection performance, total performance, and privacy protection constraints, as shown in Table 6.

The results in Table 6 show that CPRDIFL was the only method that could span multiple OSPs and achieve satisfactory CPR detection performance with data privacy protection constraints. Moreover, CPRDIFL was superior to the other methods in terms of single OSP or domain detection and overall detection. One possible reason is that the interaction and cooperation between multiple OSPs helps CPRDIFL to improve the CPR detection performance.

4.4.6. *Experiment 6: Effect of the ratio of rumor to non-rumor on CPRDIFL*

Because we used a balanced dataset in previous experiments, this may bias the actual performance of CPRDIFL, which is caused by an imbalanced class distribution [13]. To address this concern, we conducted a series of experiments on datasets with varying ratios of rumor to non-rumor, with the ratio set to 1:1, 1:2, 1:5, 1:10, 1:20, and 1:50. Fig. 6 shows the detection performance of CPRDIFL. As shown in Fig. 6, the experimental results indicate that the performance of CPRDIFL decreased significantly as the ratio decreased, which could be observed on both homogeneous and heterogeneous datasets; the accuracy, F1 score, recall, and precision values all showed a similar pattern. However, despite the performance degradation, CPRDIFL remained reasonably robust across different data distributions, even under the extreme distribution scenario, demonstrating its ability to adapt to real-world scenarios.

Table 6
Results of comparison among different models.

Rumor types	Methods	Data privacy protection status	Single F1	Total F1
Single-OSP	BMR [50]	×	92.60 %	92.60 %
Multi-domain	MDFEND [45]	×	90.01 %	91.37 %
Cross-domain	M^3FEND [51]	×	90.43 %	92.16 %
Cross-OSP	CPRDIFL (ours)	√	**95.28 %**	**95.28 %**

Note: Single F1 is used to evaluate the model performance in a single OSP or domain, and total F1 is used to evaluate the model performance in multiple OSPs or domains, reflecting the overall model performance.

5. Discussion

5.1. *Theoretical implications*

Our study offers theoretical implications in the following three aspects within the domain of CPR detection and its corresponding decision support. First, the proposed CPRDIFL framework pioneers the application of FL to CPR detection, conceptualizing it as a "federated inference" problem. This enables collaborative CPR detection leveraging complementary strengths across OSPs under data privacy protection constraints, driving a paradigm shift in rumor detection methodology. It addresses critical limitations in existing single-OSP and multi-OSP data integration approaches when confronting rumors that spread across OSPs, thereby filling a significant research gap in this domain. Second, the improved dynamic parameter aggregation strategy of CPRDIFL integrated with MacBERT simultaneously balances the differences in the data and detection capabilities of multiple OSPs. This effectively adapts to the dynamically evolving characteristics of cross-OSP rumor dissemination, substantially improving generalization performance of CPRDIFL in dynamic rumor spread environments. Finally, given the mathematical isomorphism between the distributed model training paradigm of FL and the preference aggregation mechanism in group decision-making, and considering that CPRDIFL builds upon the collaborative participation of multiple OSPs in rumor detection, the design of CPRDIFL deepens the understanding of intelligent group decision-making in privacy-sensitive and data-constrained environments. It thus explores viable pathways toward constructing intelligent group decision-making frameworks under data privacy protection conditions and advancing the formation of a new paradigm for intelligent decision science in the "Privacy Computing Era".

5.2. *Practical implications*

Our study offers three practical implications for CPR detection. First, CPRDIFL leverages the complementary strengths of diverse OSPs without requiring data integration and sharing, providing a practical solution for CPR detection. This alleviates OSPs' growing concerns regarding privacy protection, thereby facilitating the development of distributed and scalable rumor detection and regulatory decision support systems. Second, the results of Experiment 3 demonstrate that increasing the number of general OSPs enhances CPR detection

Table 5
Generalization performance of CPRDIFL on different datasets.

Dataset	Language	F1 score	Accuracy	Recall	Precision	Data Source (Amount of data)
PHEME	English	90.73 %	90.70 %	90.71 %	90.74 %	Five events from Twitter (2167, 1809, 2020, 1840, 2807)
Twitter1516	English	81.33 %	81.77 %	81.35 %	81.32 %	Twitter 15 (2030), Twitter 16 (1578)
Covid19	English	85.85 %	86.02 %	85.90 %	85.80 %	Snopes (2080), Factcheck (2051), Poynter (2706)
Mixed	English	87.74 %	87.80 %	87.76 %	87.72 %	snopes, factcheck, poynter, Twitter15, Twitter16

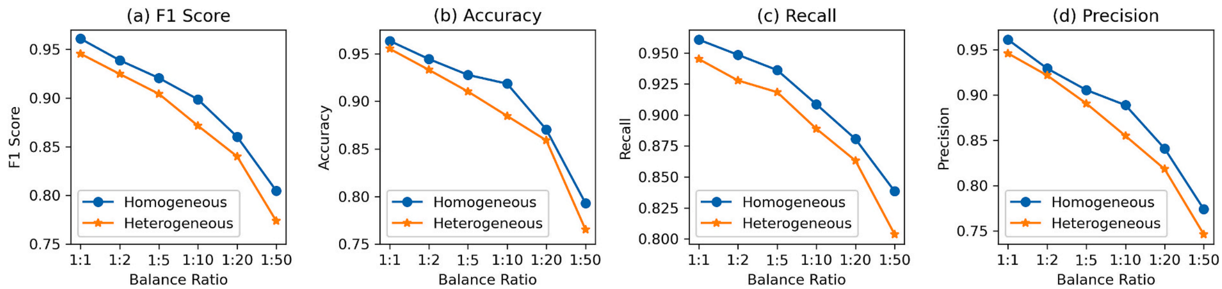

Fig. 6. Performance on different ratio of rumor to non-rumor.

performance, whereas the lack of specialized OSPs diminishes its effectiveness. This finding underscores the importance of collaborative CPR detection integrating both general and specialized OSPs. It also provides empirical support for exploring dynamic collaboration mechanisms and intervention sharing among different types of OSPs. Finally, the results of Experiment 4 indicate that CPRDIFL exhibits strong adaptability across languages and rumor events. This demonstrates its potential for practical deployment in multilingual, multi-event rumor monitoring scenarios, aiding government regulatory authorities and OSP managers in achieving accurate and timely CPR regulation within these more complex contexts.

6. Conclusions and future work

The detection of CPRs has become a significant public concern owing to the pervasive influence of social media. However, research on CPR detection is limited. To address the dual challenges of data privacy protection and the differences in the data and detection capabilities of OSPs simultaneously, we designed, implemented, and evaluated an innovative CPR detection framework named CPRDIFL. By combining the FL framework with MacBERT and introducing a dynamic parameter aggregation strategy, the dual challenges of CPR detection can be overcome, thereby achieving complementary advantages across different OSPs under the constraints of data privacy protection. Comprehensive and rigorous experimental evaluations on both homogeneous and heterogeneous datasets demonstrate that our framework significantly outperforms existing methods in single-OSP, multi-domain, and cross-domain rumor detection and effectively detects CPRs in diverse scenarios. Although this study achieved satisfactory results, it still has several limitations: (1) The CPRs considered are textual data. Because OSPs increasingly support the sharing of text, images, and videos, detecting multimodal CPRs remains a challenging task. (2) This study only performed a coarse-grained division of OSPs into general and specialized types; however, OSPs often exhibit mixed characteristics across multiple attributes. Future research could capture these platform differences at finer granularity to enhance the CPR detection performance further. (3) Although this study simulated CPRs in real scenarios as far as possible, reality is more complicated. Future research will collect multi-modal CPR information from various real OSPs to improve the detection performance further.

Declarations ethics approval and consent to participate

There are no human subjects in this article and informed consent is not applicable.

CRediT authorship contribution statement

Xuelong Chen: Writing – review & editing, Supervision, Investigation, Funding acquisition, Conceptualization. **Jinchao Pan:** Writing – review & editing, Writing – original draft, Visualization, Software, Data curation.

Consent for publication

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Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.dss.2025.114524.

Data availability

All data analyzed in this study are included in this published article.

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